

Peyyala Karthik

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TECHNICAL SKILLS

- **Languages:** C, C++, Python
- **Embedded & Electronics:** Microcontrollers, Arduino, ESP32, Sensors, Embedded Systems, PCB Design
- **Simulation & Design:** Proteus, Tinkercad, AutoCAD
- **Soft Skills:** Problem-Solving Skills, Teamwork, Communication, Adaptability

CAREER OBJECTIVE

Electronics & Communication Engineering student specializing in VLSI Design and Technology, with an interest in Embedded Systems, digital electronics, microcontrollers and automation. Interested in applying foundational technical knowledge through hands-on projects while continuously developing engineering and problem-solving skills.

PROJECTS

- **2D Memory Controller Using Decoder:** May 2026
Technologies: Proteus, 74139 Decoder, D Flip-Flops, 7447 Decoder, 7-Segment Displays
Role: Worked on the design and implementation of the digital memory-controller circuit.
Technical Responsibilities: Implemented memory selection and data-handling functions using a 74139 decoder, D flip-flops, data switches, Write Enable, 7447 decoder and 7-segment displays in Proteus.
Challenges: Coordinating the decoder, storage elements, control signals and display sections to achieve the required circuit operation.
Learning Outcome: Learned practical digital-circuit design, decoder-based memory selection, flip-flop usage, control-signal handling and circuit simulation using Proteus.
- **Water Level Indicator & Automatic Pump Control:** Apr 2026
Technologies: Arduino, Ultrasonic Sensor, Relay, LCD
Role: Contributed to building and testing the Arduino-based system.
Technical Responsibilities: Used an ultrasonic sensor to measure the distance of water in the tank, displayed the measured distance on an LCD, and used a relay to control the motor pump.
Challenges: Integrating the ultrasonic sensor, LCD, relay and motor pump into a working system.
Learning Outcome: Learned the basics of sensor interfacing, Arduino-based control, LCD interfacing and relay-based motor-pump control.
- **Obstacle Avoiding Car:** Jun 2025
Technologies: Arduino, Ultrasonic Sensor, L293D Motor Driver
Role: Contributed to assembling and implementing the obstacle-detection and movement-control system.
Technical Responsibilities: Used an ultrasonic sensor for obstacle detection and an L293D motor driver for motor control. Implemented automatic direction change when an obstacle was detected.
Challenges: Coordinating obstacle detection with motor control so that the car could respond automatically to obstacles.
Learning Outcome: Learned the basics of ultrasonic sensing, motor-driver interfacing, autonomous vehicle control and real-time response.
- **Automatic Garage Gate Control System:** Dec 2025
Technologies: PLC, Sensors
Role: Contributed to the sensor-based automatic entry and exit gate-control system.
Technical Responsibilities: Used sensors to detect vehicle entry and exit and send signals to the PLC. Implemented automatic gate opening and timed closing after the vehicle entered or exited.
Challenges: Coordinating sensor signals with PLC control and ensuring that the gates opened and closed after the required time interval.
Learning Outcome: Learned the basics of PLC-based automation, sensor-to-PLC signalling, timed control and automated gate operation.
- **Word Scramble Puzzle Game:** May 2025
Technology: Python
Role: Team member.
Technical Responsibilities: Participated in developing and presenting an interactive word-scramble puzzle game designed to help beginners practice Python through gameplay.

Challenges: Understanding and implementing the logic required for an interactive word-based game.
Learning Outcome: Gained practical exposure to Python programming and learned how programming concepts can be applied to a simple interactive application.

CERTIFICATIONS

- Cyber-Physical Systems: Sensors and PLC Interfacing | Nov 2025 [view certificate](#)
- Basics of Electrical Safety | Dec 2025 [view certificate](#)

ACTIVITIES AND PARTICIPATIONS

- **Community Development Project (CDP):** Jul 2026
Organization: Swarnandhra Seva Samstha, Rajahmundry
- **Aethera-X — 24-Hour Hackathon:** Nov 2025
Participation

EDUCATION

- **Lovely Professional University** Punjab, India
Bachelor of Technology - Electronics and communication Engineering; CGPA: 7.10
Specialization: VLSI Design and Technology 2025-present
- **Tirumala Junior College** Rajahmundry, Andhra Pradesh
Intermediate; Percentage: 94.40% April 2023 - March 2025
- **Bhashyam English Medium High School** Rajahmundry, Andhra Pradesh
Matriculation; Percentage: 85% April 2022 - March 2023